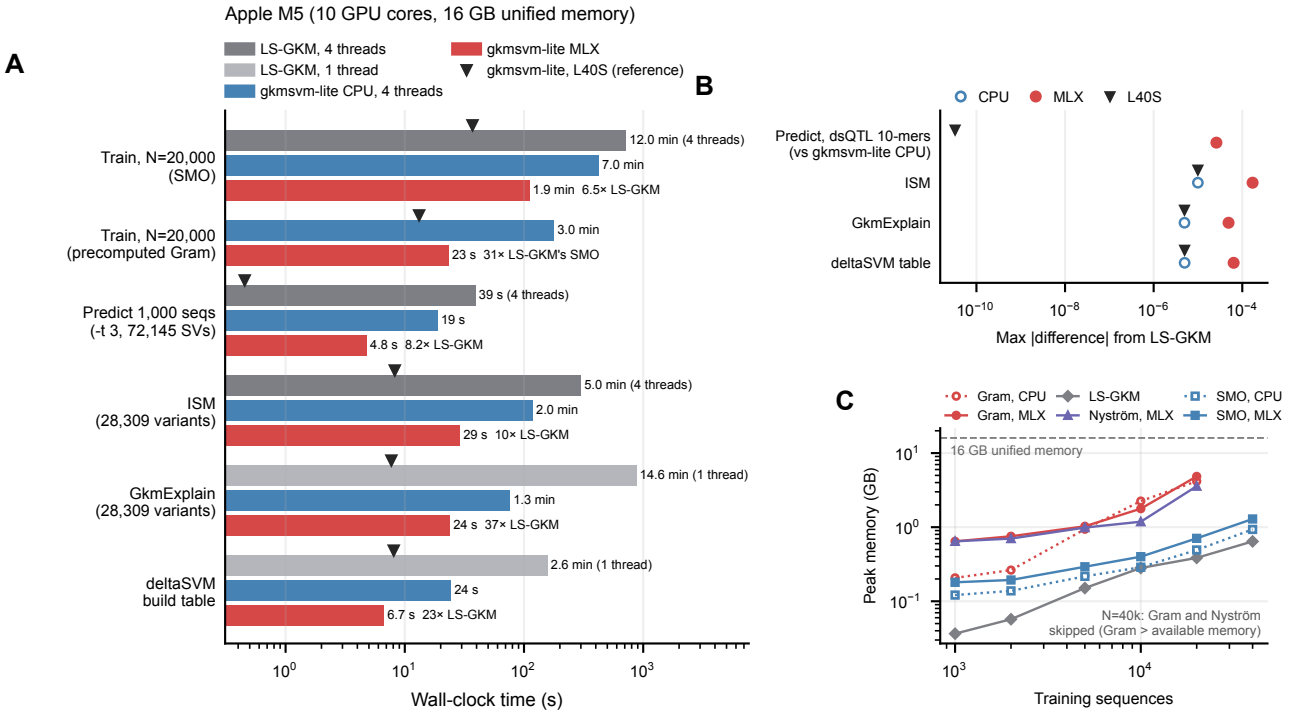
A
Apple M5 (10 GPU cores, 16 GB unified memory)
LS-GKM, 4 threads
LS-GKM, 1 thread
gkmsvm-lite CPU, 4 threads
gkmsvm-lite MLX
gkmsvm-lite, L40S (reference)
Train, N=20,000 (SMO)
12.0 min (4 threads)
7.0 min
1.9 min 6.5× LS-GKM
Train, N=20,000 (precomputed Gram)
3.0 min
23 s 31× LS-GKM's SMO
Predict 1,000 seqs (-t 3, 72,145 SVs)
39 s (4 threads)
19 s
4.8 s 8.2× LS-GKM
ISM (28,309 variants)
5.0 min (4 threads)
2.0 min
29 s 10× LS-GKM
GkmExplain (28,309 variants)
14.6 min (1 thread)
1.3 min
24 s 37× LS-GKM
deltaSVM build table
2.6 min (1 thread)
24 s
6.7 s 23× LS-GKM
10^0
10^1
10^2
10^3
Wall-clock time (s)
B
CPU
MLX
L40S
Predict, dsQTL 10-mers (vs gkmsvm-lite CPU)
ISM
GkmExplain
deltaSVM table
10^{-10}
10^{-8}
10^{-6}
10^{-4}
Max |difference| from LS-GKM
C
Gram, CPU
Gram, MLX
LS-GKM
Nyström, MLX
SMO, CPU
SMO, MLX
16 GB unified memory
Peak memory (GB)
10^1
10^0
10^{-1}
N=40k: Gram and Nyström skipped (Gram > available memory)
10^3
10^4
Training sequences